

Jinrui (Jerry) Gou

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RESEARCH INTERESTS

Information Retrieval, Search Efficiency, Learned Index Optimization, Nearest Neighbor Search

EDUCATION	New York University	Brooklyn, NY
	Ph.D. in Computer Science	09/2021-Present
	Advisor: Professor Torsten Suel	
	Cumulative GPA: 3.97/4.0	
	M.S. in Computer Science	09/2019-05/2021
	Cumulative GPA: 4.0/4.0	
	Sun Yat-Sen University	Guangzhou, China
	B.E. in Computer Science & Technology	09/2015-06/2019
	Cumulative GPA: 90/100 (3.8/4.0)	
	Mathematics Courses GPA: 95/100	

KNOWLEDGE

- **Languages:** C/C++, Python, SQL, AMPL/CPLEX, Latex, MATLAB, Bash, HTML5.
- **Tools:** Scikit-learn, PyTorch, TensorFlow, Transformers, XGBoost, Numpy, Matplotlib, CUDA.
- **Courses Enrolled:** Computer Architecture, Natural Language Processing, Machine Learning (Rank 1st in Class of 43 students), Deep Learning, Web Search Engines, Design & Analysis of Algorithms (Rank 1st in Class of 47 students), Network Design and Algorithms (Rank 2nd in Class of 20 students), Algorithmic Machine Learning and Data Science.

RESEARCH EXPERIENCE

Ph.D. Research Assistant for Prof. Torsten Suel, NYU 09/2021-Present

1. Scalable Candidate Generation and Threshold Estimation for Fast Top-K Information Retrieval

- Develop efficient early termination algorithms for ranking scoring functions, enabling fast top- k retrieval over large-scale document collections.
- Design specialized prefix index structures and fine-tune storage layout for cache-efficient access during ranking.
- Perform fast candidate lookups using highly optimized index structures to improve ranking quality and recall.
- Implement end-to-end retrieval and ranking pipeline in C++ evaluated across multiple corpora and ranking models.
- Achieve 5–10 $\times$ speedup with negligible end-to-end quality loss compared to optimized baseline algorithms, with direct applicability to multi-cascading ranking pipelines.

2. Graph-Structured Models for Approximate Nearest Neighbor Search and Scalable Retrieval

- Study the theoretical limits of constructing sparse navigable graphs for nearest neighbor search in high-dimensional embedding spaces, directly applicable to document retrieval in large-scale search systems.
- Design efficient randomized and deterministic graph construction algorithms achieving $O(\sqrt{n} \log(n))$ average degree for any point set and distance function.
- Bound the maximum degree and average degree of navigable graphs using mathematical proof.
- Prove nearly matching lower bounds showing $\Omega(n^{1/2})$ average degree is necessary for navigable graphs on random point sets.
- Develop efficient heuristic graph pruning algorithms to drastically reduce the average degree of a graph while maintaining navigability.
- Propose Distance Adaptive Beam Search, with a smarter termination condition that replaces the standard fixed beam width, improving search efficiency on graph-structured indexes.
- Prove guaranteed search accuracy on navigable graphs and achieve 30–40% fewer distance computations over state-of-the-art graph indexes (HNSW, Vamana).

3. Sparse Learned Index Structures Optimization with Large Language Models

- Contribute to a project combining LLM-based document expansion (Llama 2) with Transformer-based term importance prediction, enhanced by hard negatives and distillation.
- Assist in dataset preparation and ensure experiment reproducibility across benchmark collections.
- Conduct evaluation experiments using PISA search engine toolkit and contribute to project presentation materials.

Graduate Research Assistant for Prof. Yong Liu, Networked Systems Lab, NYU

09/2019-05/2021

1. Realtime Mobile Network Prediction for High QoS

- Collect mobile 4G/LTE bandwidth traces in metro area of NYC.
- Develop LSTM RNN models for high-accuracy realtime bandwidth prediction.
- Develop Gradient Boosting models for handoffs predictions between 4G and 5G access modes.

2. Joint Traffic Routing and Computation Server Placement in Edge Cloud Networks

- Build quasi-convex optimization model to balance traffic and computation resource allocation.
- Use AMPL/CPLEX and Python optimization packages to obtain optimal feasible solutions.

PUBLICATIONS

1. Yousef Al-Jazzazi*, Haya Diwan*, **Jinrui Gou***, Cameron Musco*, Christopher Musco*, Torsten Suel*, “Distance Adaptive Beam Search for Provably Accurate Graph-Based Nearest Neighbor Search”, *NeurIPS 2025: Advances in Neural Information Processing Systems, December 2025*¹
2. **Jinrui Gou**, Antonio Mallia, Yifan Liu, Minghao Shao, and Torsten Suel, “Fast and Effective Early Termination for Simple Ranking Functions”, *ACM SIGIR Conference on Research and Development in Information Retrieval, July 2025*.

¹All authors contributed equally.

3. Haya Diwan*, **Jinrui Gou***, Cameron Musco*, Christopher Musco*, and Torsten Suel*, “Navigable Graphs for High-Dimensional Nearest Neighbor Search: Constructions and Limits”, *NeurIPS 2024: Conference on Neural Information Processing Systems, December 2024*.²
4. **Jinrui Gou**, Yifan Liu, Minghao Shao, and Torsten Suel, “Beyond Quantile Methods: Improved Top-K Threshold Estimation for Traditional and Learned Sparse Indexes”, *IEEE International Conference on Big Data, December 2024*.
5. Soyuj Basnet, **Jinrui Gou**, Antonio Mallia, and Torsten Suel, “DeeperImpact: Optimizing Sparse Learned Index Structures”, in *ReNeuIR at SIGIR 2024: The Third Workshop on Reaching Efficiency in Neural Information Retrieval, July 2024*.
6. Lifan Mei, **Jinrui Gou**, Jingrui Yang, Yujin Cai, and Yong Liu, “On Routing Optimization in Networks with Embedded Computational Services”, *IEEE Transactions on Network and Service Management, September 2024*.
7. Lifan Mei, **Jinrui Gou**, Yujin Cai, Houwei Cao, and Yong Liu, “Realtime mobile bandwidth and handoff predictions in 4G/5G networks”, *Computer Networks, February 2022*.

EMPLOYMENT EXPERIENCE

Applied Scientist Intern, Amazon, Sunnyvale, CA

Summer 2026

- Work on applied machine learning and large-scale search systems.
- Explore efficient model serving and retrieval components under practical latency constraints.

Quantitative Research Intern, Cantor Fitzgerald, New York

Summer 2025

- Develop short-term stock price movement prediction models (10s, 20s, and 30s horizons) using time series analysis and ensemble methods including Random Forest and customized XGBoost models.
- Utilize kdb+/q for scalable preprocessing of high-frequency time series market data, performing feature engineering including joins, alignment, and technical indicators (e.g., MACD, KDJ).
- Incorporate proprietary dark pool trading data as additional predictive features, building end-to-end ML pipelines for model training and evaluation to enhance prediction accuracy.

Course Assistant, Web Search Engines & Principles of Database Systems, NYU

Spring/Fall 2025

- Design homework problems, conduct demo sessions, and grade students’ assignments and exams.
- Assist the professor by answering students’ questions on Piazza and holding weekly TA office hours.

Teaching Assistant, ECE-GY 9343 Data Structures and Algorithms, NYU

Spring–Fall 2020

Design homework problems, grade assignments and exams, and hold weekly office hours.

REFERENCES

Prof. Torsten Suel
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²All authors contributed equally.